

Current and Resistance

1 Introduction

1.1 Definitions

The electric current in a wire is defined as

$$I = \frac{dQ}{dt}$$

where dQ is the total amount of charge that passes through a cross-section of the wire in a differential amount of time, dt .

If q is the charge (in Coulombs) of each flowing charge, n is their number per volume (“number density”), v_d their average speed along the wire (called the “drift speed”), then

$I = n|q|v_dA$, where A is the cross-sectional area of the wire.

The average current density is defined as

$$J = \frac{I}{A}, \text{ which can be written as } J = n|q|v_d.$$

1.2 Ohm’s Law

If an electric field exists in a wire (by, for example, connecting its ends to a battery), the charges will accelerate until they collide with another particle and decelerate (collisions resist the flow). The net result will be current – a flow of charges with an average drift speed. Experimentally, it has been shown that in many materials, the electric field is proportional to the current density:

$$E = \rho J$$

where the value of the proportionality constant ρ , called resistivity, depends on the material. This is one version of Ohm’s law.

For a wire of length L and constant cross-sectional area A , Ohm’s law can also be written as $V = (\rho L/A)I$. If we define resistance as $R = \rho L/A$, then we have another relationship that is also referred to as Ohm’s law:

$$V = IR.$$

In this form, the interpretation is that the voltage between the ends of a wire is proportional to the current in the wire, with the proportionality constant of R .

2 Problems

2.1 Definitions and Ohm's Law

A 9-volt power source is connected to a wire of length 10 meters with a circular cross-section and radius of 0.01 meters. The wire has a resistivity of $10^{-8} \Omega \cdot \text{m}$. The number density of flowing charges is $10^{28}/\text{m}^3$. Assume Ohm's law applies and the charges that flow are electrons.

1. What is the resistance (with units) of the wire?

Answer: $R = \frac{\rho L}{A} = \frac{(10^{-8} \Omega \cdot \text{m})(10 \text{ m})}{\pi(0.01 \text{ m})^2} = \frac{10^{-3}}{\pi} \Omega$. When solving circuit problems, the resistors involved typically have a much larger resistance than this, which is why we neglect the resistance of the wires.

2. What is the current (with units) in the wire?

Answer: $I = \frac{9 \text{ V}}{\frac{10^{-3}}{\pi} \Omega} = 9,000\pi \text{ A}$. This is a huge current. If you look at the back of an electronic device, you will see a rating on the order of 1 A. Household circuit breakers are set to open ("trip") at approximately 15 A. Most power sources cannot supply current at this rate; even if the power source could supply this current, the amount of heat created would lead to a fire or melting of the wire.

3. How much charge (in Coulombs) flows past a cross-section of the wire per second?

Answer: $9,000\pi \text{ C}$

4. How many electrons flow past a cross-section of the wire per second? (The charge on an electron is $-1.6 \cdot 10^{-19} \text{ C}$.)

Answer:

$$9,000\pi \text{ C} \cdot \frac{1 \text{ electron}}{1.6 \cdot 10^{-19} \text{ C}} \approx 1.8 \cdot 10^{23} \text{ electrons}$$

Have you ever encountered a number on the order of 10^{23} before?

5. What is the current density (with units)?

Answer: $J = \frac{I}{A} = 9 \cdot 10^7 \frac{\text{A}}{\text{m}^2}$

6. What is the drift velocity of electrons in the wire? (The charge on an electron is $-1.6 \cdot 10^{-19} \text{ C}$.)

Answer: $v_d = J/n|q| \simeq 6 \text{ cm/s}$

7. Based on the description of how charged particles flow in a wire, explain why the resistance of a cylindrical wire is proportional to its length and inversely proportional to the square of its radius.

Answer: See the description in your textbook.

2.2 Definitions and Ohm's Law

A power source is connected to a 20 meter long wire with a circular cross-section and radius of 0.01 meters. The wire has a resistivity of $10^{-7} \Omega \cdot \text{m}$. The number density of flowing charges is $10^{27}/\text{m}^3$. The current in the wire was measured and found to be 1 A. Assume Ohm's law applies and the charges that flow are electrons.

1. What is the resistance (with units) of the wire?

Answer: $(.02/\pi) \Omega$

2. How much charge (with units) flows past a cross-section of the wire per second?

Answer: 1 C

3. What is the current density (with units)?

Answer: $J = \frac{10^4}{\pi} \frac{\text{A}}{\text{m}^2}$

4. What is the drift velocity of electrons in the wire? (The charge on an electron is $-1.6 \cdot 10^{-19}$ C.)

Answer: $v_d = J/n|q| = 0.02 \text{ mm/s}$

2.3 Current Through a Cylindrical Shell

If a cylindrical wire with an inner radius a and outer radius b carries a current I , what is J ?

Answer: $J = \frac{I}{\pi(b^2 - a^2)}$

2.4 $I = n|q|v_dA$ derivation

Derive the relationship $I = n|q|v_dA$. Provide a diagram.

Answer: See textbook.